

This Zenodo deposit contains a publicly available description of the Dataset:

"Human PD Cingulate GeoMx".

Dataset Description:

Segments were captured from specific cortical layers of Parkinson's disease cases capturing whole transcriptomes from neurons positive for Lewy bodies, or negative for Lewy bodies using the GeoMx WTA platform.

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Dataset Name: biederer-pmdbs-spatial-geomx-lamda, v0.1

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ASAP Lab:

ASAP Project: Understanding and manipulating cellular and circuit-level vulnerability to neurodegeneration in Parkinson's disease

Project Description: Persons with Parkinson's disease often experience cognitive decline and dementia that is well correlated with Lewy pathology in the neocortex. To address how PD impacts cognition, we study the impact of cellular pathology on cortical transcription, connectivity, and function. First, we determine how pathology causes the breakdown of synapses, the cellular structures that connect neurons into networks. Second, we establish links between pathology, dysregulated gene expression, and altered neuronal activity. Third, we use mathematical modeling to understand regional vulnerability. Our findings aim to reveal the molecular dysfunctions and cortical network aberrations that contribute to cognitive impairments in PD.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset